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Title	Modelling travellers' attitudes to security, surveillance and privacy across the EU
Track	General Papers
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Abstract	Following revelations of NSA secret surveillance programmes in the summer of 2013, there has been an intensified debate on privacy and personal data. While internet surveillance was the focus of this debate, it may also have far-reaching consequences in other fields, including transportation, where increasing amounts of data are collected through mass surveillance. For example, the data collected through the surveillance of metro/train facilities is becoming more effective as closed circuit television evolves to gain capabilities such as facial recognition and advanced security checks involving body scanners and pat-down also have significant potential to increase security, but both at the expense of travellers' privacy. It is likely that the security and surveillance measures of the future will be judged on their effects on privacy.

PACT — Public Perception of Security and Privacy: Assessing Knowledge, Collecting Evidence, Translating Research into Action — is a three year project, funded by the European Commission's 7th Framework Programme, that aims to understand more fully the public perception of security, privacy and surveillance across the 27 European Member States (at the time). The knowledge gained will be used to inform the development of a new reference architecture for privacy and a decision support system.

PACT employs a quantitative survey, which includes stated preference (SP) methods to collect the security, surveillance and privacy related preferences using face-to-face and internet-based surveys for over 26,000 respondents. Three SP experiments have been designed and undertaken to investigate security, surveillance and privacy related preferences in travel, health and internet contexts. In addition to the stated preference experiments, the survey collects attitudinal information related to security, privacy, trust and risk-taking as well as socio-economic information about the respondent and their household. Thus the survey provides a rich dataset that allows the impact of attitudes to security, surveillance and privacy issues to be modelled.

The analysis presented in this paper is based on the SP data collected in the travel context in which respondents are asked to imagine that they are just about to make a journey by train or metro/underground/subway. Each SP scenario presents respondents with two hypothetical options for making that journey. Each option presents different levels and types of: surveillance (e.g. presence and type of security cameras), security personnel, security checks and time to go through security. Options also involve varying levels of a security surcharge on the ticket cost or no surcharge. A third option allows respondents to opt out from choosing either of the two travel options on offer.

Models are developed to explain the observed choices as a function of the attributes of each alternative, including cost, and the socio-economic characteristics of the individual. These models include error scales estimated by

country and survey mode (face-to-face relative to internet-based) to control for differences in variance between countries and survey methods.

Attitudinal scales are introduced into the models using two latent variables, 'privacy concern' and 'distrust'. 'Privacy concern' is measured using four statements relating to concerns while travelling in a train or metro about misuse of CCTV images by the authorities, misuse of travel data to track an individual's whereabouts, misuse of security measures for sexual or racial harassment, and sharing travel data and CCTV images across and outside the EU. Responses to each of these four statements are recorded using a five-point Likert-scale. 'Distrust' is measured using four statements introduced by Westin-Harris (see Kumaraguru and Cranor, 2005) that probe respondents' general views as to whether technology has got out of control, whether government can be trusted to look after our interests, the impact of individuals' votes on what the government does, and whether in general business helps us more than it harms us.

Attitudes are incorporated by adding the attitudinal variables to the 'opt out' option and estimating parameters expressing the relative impact on the SP choices, and by interacting the attitudinal variables with the socio-economic characteristics of the individual.

The preliminary attitudinal privacy-concern model indicates that younger travellers, higher income travellers and travellers with tertiary education have lower levels of privacy concern relative to older travellers, lower income travellers and those without tertiary education. The attitudinal-distrust model indicates that younger travellers, higher income travellers, travellers with tertiary education and workers have lower levels of distrust relative to older travellers, lower income travellers, those without tertiary education and non-workers. The model development will be completed and reported by the end of April 2014.

McFadden (1986) already argued for the incorporation of latent variables in choice models and two empirical applications of the approach were presented in Ashok *et al.* (2002). Latent variables for concern and privacy were successfully identified by Daly *et al.* (2012) who developed models using SP data collected in a UK study investigating the impact of security on rail travel behaviour. Daly *et al.* noted that many other studies in this area have not recognised the repeated choice nature of SP data; this is explicitly taken into account in the models estimation for this study. The socio-economic variables available in the Daly *et al.* study were limited, and this study builds on that work by investigating the impact on attitudes of a wider range of socio-economic variables, and in particular investigates whether attitudes vary with income and across Europe.

References:

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